

1. 解: $\dot{x} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u$

$$|sI - A| = \begin{vmatrix} s-1 & 1 & -1 \\ 0 & s-1 & -1 \\ -1 & 0 & s-1 \end{vmatrix} = (s-1)^3 + 1 - (s-1) = s^3 - 3s^2 + 2s + 1$$

$$[s - (-1)][s - (-2)][s - (-3)] = s^3 + 6s^2 + 11s + 6$$

$$k = [-5 \quad -9 \quad -9]$$

$$U = [b \quad Ab \quad A^2b] = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix}$$

$$U^{-1} = \frac{1}{|U|} \begin{bmatrix} 0 & -1 & 1 \\ 2 & -1 & 0 \\ -1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 1 \\ 2 & -1 & 0 \\ -1 & 1 & 0 \end{bmatrix}$$

$$q = [0 \quad 0 \quad 1] U^{-1} = [-1 \quad 1 \quad 0]$$

$$T^{-1} = \begin{bmatrix} q \\ qA \\ qA^2 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 2 & 0 \\ -3 & 1 & -3 \end{bmatrix}$$

$$K' = K T^{-1} = [23 \quad -50 \quad -9] \text{ 为反馈增益向量}$$

$$u = [23 \quad -50 \quad -9] x + r$$

2. 解: $\dot{x} = \begin{pmatrix} 1 & -2 & -2 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix} x + \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} u$

判断能控性

$$M = [b \quad Ab \quad A^2b] = \begin{bmatrix} 2 & -4 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & -5 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -5 \end{bmatrix}$$

$$\text{rank}(M) = 3 = n$$

$\therefore$ 系统^{完全}能控

$\therefore$ 系统可以通过状态反馈加以镇定

$$3. \quad (1) \quad |sI - A| = \begin{vmatrix} s & 1 & 0 & 0 \\ 0 & s & 1 & 0 \\ 0 & 0 & s & -1 \\ 0 & 0 & 1 & s \end{vmatrix} = s^4 - 11s^2 = 0$$

$$s_{1,2} = 0 \quad s_{3,4} = \pm \sqrt{11}$$

A的特征根中有 $s = \sqrt{11}$ 正实部特征根，因此系统开环不稳定

(2) 判断能控性

$$B = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad B_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}$$

$$M = [B \quad AB \quad A^2B \quad A^3B]$$

$$= \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & -11 \\ -1 & 0 & -11 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -10 \\ 0 & 0 & -10 & 0 \end{bmatrix}$$

$$\text{rank}(M) = 4 = n$$

$\therefore$ 系统能控 完全能控

$\therefore$ 系统可以镇定